

Udipta Mondal

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Education

Presidency University Bangladesh

Jan 2022 – Dec 2025

BSc in Computer Science and Engineering

Dhaka, Bangladesh

CGPA: 3.10 | Thesis: ML-Based Predictive Modeling for Early Detection of Dengue Disease

Technical Skills

Languages:	Python, C++
AI & LLMs:	CrewAI, Gemini API, Prompt Engineering, RAG Pipelines, Vector Databases
ML & DL:	Machine Learning, Deep Learning, TensorFlow, Keras, scikit-learn, NumPy, Pandas, MLflow
Computer Vision:	OpenCV, YOLOv11, ByteTrack, Object Detection & Tracking, MediaPipe
Backend & Dev:	FastAPI, Docker
Tools:	Git/GitHub, Jupyter, Kaggle, VS Code, Streamlit, Render
Concepts:	OOP, Competitive Programming

Soft Skills

Analytical problem-solving and algorithmic thinking • Team collaboration, communication, and leadership • Competitive programming

Projects

AI Tutor Platform: Multimodal Adaptive Learning Engine *RAG, ChromaDB, Gemini API, MLflow, scikit-learn*

- Architected a personalized AI tutoring system where students upload study material and receive Socratic-style guidance powered by a hybrid RAG pipeline (ChromaDB vector store + Gemini API); [live demo](#).
- Built a multimodal grading engine that applies OCR to evaluate handwritten mathematical derivations with partial-credit scoring, coupled with a scikit-learn HistGradientBoosting model for real-time weak-topic knowledge tracing.

AegisDID: Sybil-Resistant Decentralized Identity *MediaPipe, ZK-ML, Circom, SnarkJS*

- Architected a privacy-first identity verification system for humanitarian aid — runs MediaPipe face mesh inference entirely on-device to extract 128-D facial embeddings, ensuring zero biometric data leaves the client; [live demo](#).
- Designed ZK-ML circuits in Circom/SnarkJS to prove identity without revealing raw data, and deployed nullifier-tracked Base Sepolia smart contracts to cryptographically prevent duplicate relief claims.

ResearchFlow AI: Autonomous Deep Research Engine *CrewAI, Gemini API, LLM Orchestration, Tavily*

- Designed a 4-agent autonomous research pipeline (Researcher → Analyst → Writer → Editor) using CrewAI and Tavily web search — takes a topic as input and produces a structured, cited, publication-ready report with no manual intervention.
- Delivered a Streamlit dashboard with live agent hand-off visualization, automated rate-limit retry logic, and multi-format export, reducing end-to-end research time from hours to minutes.

Falcon AI: Autonomous Border Surveillance System *YOLOv11, OpenCV, Computer Vision, Flask*

- Engineered a real-time autonomous surveillance system that runs YOLOv11 object detection on live video streams to identify and classify potential border threats with low-latency inference.
- Processed detection outputs through a tactical analysis module to generate prioritized, actionable intelligence alerts delivered directly to frontier security personnel.

People Flow Detection & Spatial Heatmap Analytics *YOLO11, ByteTrack, OpenCV, Computer Vision*

- Developed a real-time pedestrian analytics system combining YOLO11n detection with ByteTrack multi-object tracking to maintain persistent identity across dense, occluded video sequences.
- Implemented bidirectional virtual-line crossing counters for accurate in/out flow measurement, and overlaid 2D Gaussian density heatmaps to expose spatial congestion hotspots at a glance.

CardioPredict: Heart Disease Clinical Risk Engine *Random Forest, scikit-learn, FastAPI, Docker*

- Trained and tuned a Random Forest classifier across 13 clinical biomarkers (cholesterol, resting BP, age, ECG readings, etc.), achieving **83.33% accuracy** and **93.97% ROC-AUC** on CHD risk stratification.
- Productionized the model as an async FastAPI service with Pydantic schema validation, an interactive risk-scoring dashboard, and a multi-stage Docker build deployed on Render.

Awards & Achievements

- Rank 8** — Blockchain Olympiad Bangladesh, AI Category (2025) [cert]
- Nominated** — International Data Science Olympiad, Hong Kong (2025)
- ICPC Asia Dhaka Regional Contest Participant (2023) [cert]
- Winner** (Season 2) & **1st Runner-Up** (Season 1) — Presidency Intra-University Programming Contest
- 2nd Runner-Up** — AI Contest, Presidency University Bangladesh (2024)